

01 | OVERVIEW



Exploring Coastal Geography: Pile Bay, Barbados

Abstract

Pile Bay is a small fishing village on the west coast of Barbados, where every home lies within 450 meters of the sea. Researchers from Northeastern University's Global Resilience Institute and the University of the West Indies worked with residents to map the community's strengths, its exposure to storms and sea-level rise, and the everyday barriers that make recovery harder. This lesson uses their StoryMap to explore how geography shapes climate risk.

Introduction

Small Island Developing States face compounding pressures from climate change, biodiversity loss, pollution, and natural hazards. Barbados is a low-lying coral island whose highest point is only 343 meters above sea level. About a quarter of its 280,000 people live within 2 kilometers of the western coast. Pile Bay, home to just over 80 residents, sits between the busy Spring Garden Highway and the shoreline, about 5 kilometers north of Bridgetown. Its beach supports traditional fishing, and its neighbors include a power plant, a rum distillery, and a gas depot. Through the Advancing Community Climate Resilience Project (ACCRP), researchers asked a key question: how can a coastal community prepare for storms it has not seen in generations?

How Do Scientists Study This?

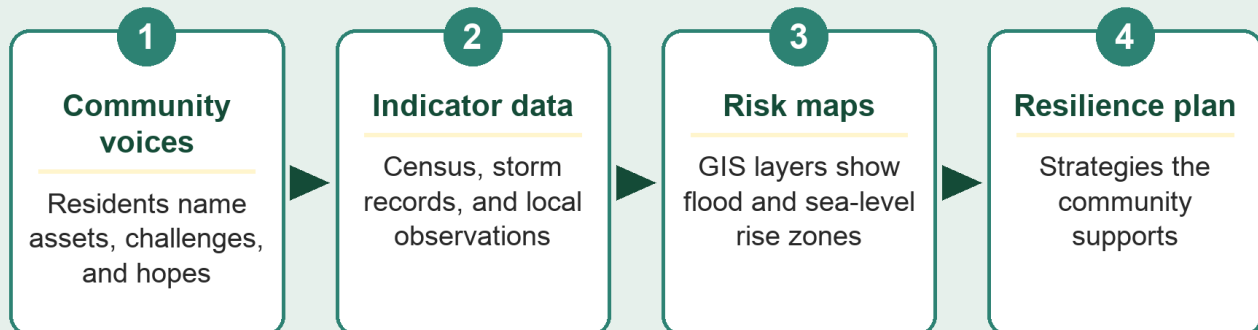


FROM THE ARTICLE

Fishing boats on a Barbados beach.
Image: ACCRP StoryMap, Global Resilience Institute.

The research team used the Integrated Resilience Enhancement System (I-RES), an assessment tool that combines community voices with measurable indicators. In July 2024, residents took part in co-development sessions where they identified community assets in four domains: physical, economic, governance, and social. Analysts combined these conversations with census data, historical storm records, and geographic information systems (GIS). They built a flood-risk map from digital elevation, land use, and distance to waterways, and a sea-level rise map showing which areas would flood under an extreme scenario. The team used a bottom-up and top-down approach, pairing local knowledge with national data so that the results would be both accurate and meaningful to residents.

How the I-RES assessment works



Study the Process

Listen first

Residents shared their assets, concerns, and hopes in co-development sessions held in Pile Bay in July 2024.

Map the hazards

Analysts used elevation, land use, and waterway data in GIS to map flood and sea-level rise risk.

Connect the evidence

Indicators from four resilience domains were combined to show where the community is strong and where it is vulnerable.

What Did They Discover?

Pile Bay's coastal location, low elevation, and flat, paved terrain put it at high risk of flooding. Storm surges already reach 150 to 300 meters inland, and sea levels in the Caribbean are projected to rise 0.5 to 1.4 meters by 2100. Many homes, the power plant, and the upgraded fishing facility are within 30 meters of the shore, and some residents cannot get home insurance at all. There is only one road in and out, and it floods often. Because Barbados has not suffered a catastrophic hurricane since Hurricane Janet in 1955, researchers identified preparedness apathy as a key challenge. Meanwhile, coral reefs that buffer storms have declined, with a 2005 bleaching event killing 25% of live coral.

Pile Bay at a glance

450 m

the whole community lies within this distance of the coast

1.4 m

Caribbean sea-level rise possible by 2100

1955

last catastrophic hurricane: Hurricane Janet

Residents said they would only consider relocating if the whole community moved together, showing how a strong sense of place shapes resilience decisions.

Takeaways

Geography sets risk

Low, flat land close to the sea makes Pile Bay highly exposed to floods, storm surge, and erosion.

Memory matters

Decades without a major hurricane can create preparedness apathy, even as climate risks grow.

Community is an asset

Fishing heritage, local volunteers, and social ties are strengths to build resilience on.

Key Terms

Small Island Developing States (SIDS)

A group of island nations that share challenges such as small size, remoteness, and high exposure to climate change and natural hazards.

Storm surge

A rise in sea level caused by a storm's winds pushing water onshore, which can flood coastal land far beyond the normal tide line.

Co-development

A research approach in which scientists and community members work together to define problems and design solutions.

GIS (geographic information system)

Computer mapping software that layers data, such as elevation and land use, to show patterns across a place.

Key Terms continued

Preparedness apathy

A loss of motivation to prepare for disasters, often after many years without a serious event.

Activity

Map Your Community's Risk

Students build a simple hazard and asset map of Pile Bay, then apply the same thinking to their own neighborhood.

Materials: Printed outline map of Pile Bay or a satellite view; Colored pencils or digital markers; Sticky notes

1. Mark the coastline, the Spring Garden Highway, and the single road in and out of Pile Bay.
2. Shade the area within 450 meters of the coast and add the key buildings named in the lesson: homes, the fishing facility, and the power plant.
3. Use sticky notes to label assets in four domains: physical, economic, governance, and social.
4. Identify two places where hazards and assets overlap and explain why that matters during a storm.
5. Reflect: what is one asset in your own community that would help people recover from a disaster?

References

Global Resilience Institute, Northeastern University, and University of the West Indies. Advancing Community Climate Resilience Project: Pile Bay, Barbados (ArcGIS StoryMap).
<https://storymaps.arcgis.com/stories/6dccd4993a1d4b01a07fd4628ce84c71>